VERTITY AI -- ROUND 3 FINAL PRODUCT DOSSIER

Autonomous Pre-Publication Brand Governance & Human-in-the-Loop Editorial Cockpit

Inkloom AI Hackathon (SIES GST) | Team LocalHost-Lunatics | Final Round 3 Submission

VERIFIED SUBMISSION LINKS & PROJECT REPOSITORY

Live Cloud Deployment: https://vertity-ai.onrender.com/

GitHub Source Repository: https://github.com/siddharthxploits69-code/Localhost_lunatics.git

`API Docs & Health Telemetry: https://vertity-ai.onrender.com/docs | /api/health`

1. EXECUTIVE PRODUCT SUMMARY

In modern multi-channel marketing, brand drift causes irreversible customer confusion, legal liability, and statutory non-compliance. Outdated pricing, unverified superlatives, off-brand color schemes, missing accessibility contrast, and undisclosed AI content slip through fragmented communication channels.

Vertity AI solves this by acting as an autonomous pre-publication brand governance gateway. It inspects copy, uploaded banners, and live drafts against verified enterprise knowledge bases. Unlike black-box checkers, every issue cites exact corporate policies and provides intent-preserving 1-click fixes with human sign-off.

2. 1-CLICK DEMO PERSONAS FOR EVALUATION

Judges can switch roles instantly in the header bar without manual login or registration friction:

Brand Admin (admin@vertity.ai)	Full access: Manage Rulebook, upload policy PDFs, view analytics.
Agency Creator (creator@agency.com)	Creator Studio: Audit drafts, inspect visual banners, accept 1-click fixes.
Enterprise Reviewer (reviewer@enterprise.com)	Review Cockpit: Editorial queue, approve drafts, request revisions with notes.

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THE THREE ROUND 3 IMPROVEMENTS DELIVERED

IMPROVEMENT 1: MULTIMODAL VISUAL BANNER & WCAG 2.1 AA INSPECTOR

`Endpoint/Source: /api/audit-visual (Pillow Quantization + W3C Luminance + Gemini Vision)`

- Pixel Palette Quantization: Extracts actual dominant HEX/RGB swatches from image bytes.
- W3C WCAG 2.1 AA Math: Computes exact relative luminance (L = 0.2126R + 0.7152G + 0.0722B).
- Verifies minimum 4.5:1 text-to-background contrast ratio for accessibility compliance.
- Gemini Multimodal Inspection: Scans for brand logo presence, OCR text drift, and auto-generates alt-text.
- Interactive Bounding Heatmaps: Visual overlays show coordinates of detected non-compliant areas.

IMPROVEMENT 2: LIVE BROWSER EXTENSION & IN-APP CMS COPILOT

`Endpoint/Source: chrome_extension/ & /api/audit-inline & /api/extension/download`

- Manifest V3 Extension: Ready for 1-click download from /api/extension/download for Chrome/Edge.
- Live In-Line Linting: Fast non-persisting endpoint (/api/audit-inline) powers real-time underlines.
- Longest-Match Non-Overlapping Spans: Prevents duplicate tags and ensures clean 1-click replacements.
- Multi-Platform Support: Seamlessly lints Google Docs, LinkedIn, WordPress, and enterprise CMS editors.
- Built-in Interactive Simulator: Test extension workflows directly inside the web UI without setup.

IMPROVEMENT 3: EU AI ACT (ART. 50) COMPLIANCE & CRYPTOGRAPHIC PDF RECEIPTS

`Endpoint/Source: pdf_generator.py & /api/receipts/{sub_id}/pdf & /api/receipts/{hash}/verify`

- EU AI Act Article 50: Verifies statutory transparency for synthetic or generative AI marketing assets.
- 1-Click Disclaimer Injection: Automatically appends mandated legal disclosures into drafts.
- Pure-Python PDF 1.4 Engine: Zero heavy dependencies (no ReportLab/WeasyPrint), 50ms generation time.
- Tamper-Proof SHA-256 Ledger: Computes content digest, active rulebook snapshot, and master seal.
- Public Verification Endpoint: Validates cryptographic seals against the immutable ledger in real time.

DELIVERY COMMITMENT VERIFICATION

All 3 commitments declared at Round 2 are 100% completed, fully deployed on Render, and covered by the automated verification test suite (python test_api.py).

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COMPLETE FEATURE CATALOG & BRAND DRIFT PILLARS

THE 5 PILLARS OF BRAND GOVERNANCE

Pillar 1: Outdated Facts & Pricing
Catches legacy 2023 pricing ($49/mo) and replaces with approved 2025 Tier ($79/mo).

Pillar 2: Unsupported Claims
Flags unverified superlatives ('fastest in the world', '100% bug-free') and enforces proof.

Pillar 3: Tone & Terminology
Enforces strict customer terminology ('Growth Partners' instead of 'clients' or 'customers').

Pillar 4: Visual & Palette Drift
Extracts dominant RGB/HEX swatches and flags non-approved off-brand visual palettes.

Pillar 5: Digital Accessibility (WCAG)
Evaluates 4.5:1 text-to-background luminance contrast and checks for descriptive alt-text.

ENTERPRISE HUMAN-IN-THE-LOOP WORKFLOW MODULES

Creator Studio	Real-time drafting canvas with instant hybrid scoring, visual banner upload, multi-format file ingestion (.txt, .md, .csv), and 1-click intent-preserving fix acceptance.
Brand Rulebook Hub	Full RBAC rule management: Admins can dynamically add, edit, or delete rules, attach verified PDF policy excerpts, and review active rule definitions. Non-admins see read-only excerpts.
Review Cockpit	Human-in-the-Loop review queue. Brand directors can filter drafts by approval state, inspect flagged violations, issue revision requests with custom notes, or grant publishing sign-off.
Governance Insights	Executive analytics dashboard aggregating repeated drift patterns across teams, top flagged terms, compliance health distributions, and automated AI policy recommendations.

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TECHNICAL ARCHITECTURE & DATA FLOW

HYBRID DUAL-ENGINE ARCHITECTURE

1. Deterministic Rule Matcher (Fast Path, <10ms): Evaluates strict terms, obsolete pricing, and regex patterns.
2. Multimodal Gemini 2.5 Flash Engine: Evaluates nuanced claims, brand tone, visual banner composition, and OCR drift.
3. Intent-Preserving Fix Synthesizer: Proposes localized minimal corrections preserving the author's voice.
4. Regulatory & Accessibility Validator: Enforces WCAG 2.1 AA luminance formulas and EU AI Act Article 50 disclosures.
5. Cryptographic Ledger & PDF Streamer: Generates SHA-256 content hashes, active rulebook digests, and streams receipts.

COMPONENT SPECIFICATIONS

Backend Framework	FastAPI (Python 3.10+), Uvicorn ASGI server, Pydantic v2 data models.
AI / ML Engine	Google Gemini 2.5 Flash Multimodal Vision & Semantic API (google-genai SDK).
Computer Vision & Math	Pillow (PIL) pixel quantization, CIE 1931 relative luminance contrast math.
PDF Receipt Engine	Pure-Python PDF 1.4 document builder (io, zlib, hashlib). Zero external dependencies.
Browser Extension	Chrome/Edge Manifest V3, content scripts, non-overlapping span replacement.
Frontend Web App	Single-Page Application (HTML5, Tailwind CSS, Lucide Icons, Canvas Confetti).
Deployment & Cloud	Render Web Service with automated continuous deployment from GitHub.

SECURITY, PRIVACY & PRODUCTION RESILIENCY

- Non-Persisting Inline Linting: Keystroke drafts from the browser extension are never saved to database.
- Deterministic Fallback: If AI engine quotas or keys are unavailable, deterministic rules execute with 100% uptime.

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TESTING VERIFICATION, RELIABILITY & ROADMAP

AUTOMATED QUALITY VERIFICATION SUITE (7/7 PASSED)

[TEST 1] GET /api/health	**PASSED**	Online telemetry verified; server responds in <5ms.
[TEST 2] GET /api/guidelines	**PASSED**	Retrieved 9 grounded brand guidelines across all 5 drift pillars.
[TEST 3] POST /api/audit	**PASSED**	Deterministic + Gemini semantic auditing verified; score & fixes returned.
[TEST 4] GET /api/analytics	**PASSED**	Drift metrics, repeated violation terms, and policy recommendations verified.
[TEST 5] POST /api/audit-visual	**PASSED**	Pillow quantization, WCAG 4.5:1 luminance, and Gemini Vision verified.
[TEST 6] POST /api/audit-inline	**PASSED**	Browser extension live linting & ZIP packaging download verified.
[TEST 7] GET /api/receipts	**PASSED**	Cryptographic PDF receipt download (%PDF-1.4) & SHA-256 seal verified.

PRODUCT ROADMAP & FUTURE HORIZONS

- **Phase 1 (Round 2 Delivered):** Hybrid Audit Engine, Knowledge Base RAG, 1-Click Fixes, Review Cockpit.
- **Phase 2 (Round 3 Delivered):** Visual Banner/WCAG 2.1, Browser Extension/CMS, EU AI Act Art. 50 & SHA-256 PDF Receipts.
- **Phase 3 (Post-Hackathon Q1 2027):** Native Figma & Canva plugins for live canvas linting inside designer tools.
- **Phase 4 (Enterprise Q2 2027):** Multi-lingual brand governance supporting 24 European and Asian languages.
- **Phase 5 (RegTech Q3 2027):** Decentralized cryptographic ledger anchor (Polygon/Ethereum) for public multi-enterprise audits.

FINAL TEAM SIGN-OFF -- TEAM LOCALHOST-LUNATICS (SIES GST)

All features, improvements, automated test cases, and cloud deployments have been completed within the strict 72-hour event window for Inkloom AI Hackathon Round 3. Live on Render & GitHub.

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